

EXPERIMENT REPORT

TITLE GENERATION ALGORITHM -
EXPERIMENT CYCLE 1

Observations and Analysis - Ready for Iteration 2

EXPERIMENT-20260114_112714

EXPERIMENT:	Title Generation Algorithm
CYCLE:	1
DATE:	2026-01-14 11:27:14
STATUS:	Observations Complete
NEXT:	Iteration 2

⚠ INFO

EXPERIMENTAL FINDINGS - Ready for next iteration cycle

ITERATIVE IMPROVEMENT PROCESS

Title Generation Algorithm - Experiment Observations

Experiment Date: 2026-01-14
Iteration: Cycle 1
Status: Observations Recorded - Ready for Iteration 2

EXPERIMENT SETUP

Initial Conditions

- **Starting Point:** Title generation was truncating claims to 60 characters
- **Problem:** Titles were not informative and got cut off
- **Goal:** Create informative titles that convey useful information

Changes Made

1. Created `generate_informative_title()` function
2. Implemented pattern matching for subject/action extraction
3. Improved CSS for multi-line title display
4. Added handling for parentheses, commands, and technical terms

TEST RESULTS: 12 VARIATIONS

Variation 1: Commands in Parentheses

Claim: The feedback system (`/love-you` and `/hate-this` commands) has been implemented
Title: feedback system (`/love-you` and `/hate-this` commands) (PROVEN)
Length: 60 chars
Assessment:  Excellent - Preserves key information (commands)

Variation 2: Technical Details with Parentheses

Claim: The PDF generator footer now displays AI assistant information (model name and generation date) instead of arXiv 2026
Title: PDF generator footer: AI assistant (model name and generation date)
Length: 67 chars
Assessment:  Good - Preserves technical details and parentheses

Variation 3: Simple Improvement ⚠️

Claim: The title generation algorithm has been improved to create more informative titles

Title: title generation algorithm: improved (PROVEN)

Length: 45 chars

Assessment: ⚠️ Loses context - "improved" is vague, doesn't say what was improved

Variation 4: Multiple Features ⚠️

Claim: The user authentication system now supports OAuth2 and JWT tokens

Title: user authentication: OAuth2 (PROVEN)

Length: 36 chars

Assessment: ⚠️ Loses JWT tokens - Only shows first feature

Variation 5: Long Action Description ❌

Claim: The database migration script successfully converts legacy data to the new schema

Title: database migration script successfully converts legacy data to (PROVEN)

Length: 71 chars

Assessment: ❌ Truncated - Cuts off "new schema" which is key information

Variation 6: API Endpoint ✅

Claim: The API endpoint /api/v1/users returns user data in JSON format with pagination

Title: API endpoint /api/v1/users: user data JSON (PROVEN)

Length: 51 chars

Assessment: ✅ Good - Preserves endpoint and key format info

Variation 7: Mobile Devices ⚠️

Claim: The frontend component renders correctly on mobile devices (iOS and Android)

Title: frontend: correctly mobile devices (iOS and Android) (PROVEN)

Length: 61 chars

Assessment: ⚠️ Awkward phrasing - "correctly mobile devices" is grammatically odd

Variation 8: Long Description ❌

Claim: The error handling mechanism logs all exceptions to a centralized logging service

Title: error handling mechanism logs all exceptions to a centralized (PROVEN)

Length: 70 chars

Assessment: ❌ Truncated - Cuts off "logging service" which is important

Variation 9: Percentage Metrics

Claim: The test suite coverage has increased from 60% to 85% across all modules

Title: test suite coverage: increased (PROVEN)

Length: 39 chars

Assessment:  Loses metrics - "60% to 85%" is important information

Variation 10: Long Pipeline Description

Claim: The deployment pipeline automatically builds and deploys to production when tests pass

Title: deployment pipeline automatically builds and deploys to production (PROVEN)

Length: 75 chars

Assessment:  At limit - Loses "when tests pass" condition

Variation 11: Performance Metrics

Claim: The caching layer reduces database query time by 70% for frequently accessed data

Title: caching layer reduces database query time by 70% for frequently (PROVEN)

Length: 72 chars

Assessment:  Truncated - Loses "accessed data" but keeps key metric (70%)

Variation 12: Security Features

Claim: The security module now validates all input data and prevents SQL injection attacks

Title: security module now validates all input data and prevents SQL (PROVEN)

Length: 70 chars

Assessment:  Truncated - Cuts off "injection attacks" but keeps "SQL" which hints at it

KEY OBSERVATIONS

What Works Well

1. **Commands/Features in Parentheses:** Excellent preservation (Variations 1, 2, 7)
2. **Technical Terms:** Good handling of AI, API, PDF, JSON (Variations 2, 6)
3. **Short, Focused Claims:** Works well for concise claims
4. **Endpoint Preservation:** API endpoints are well-preserved (Variation 6)

Areas Needing Improvement

1. **Long Action Descriptions:** Truncates before key information (Variations 5, 8, 10, 11, 12)

2. **Multiple Features:** Only shows first feature, loses others (Variation 4)
3. **Metrics/Percentages:** Loses important quantitative data (Variation 9)
4. **Grammatical Flow:** Sometimes creates awkward phrasing (Variation 7)
5. **Vague Outcomes:** "improved" is too generic (Variation 3)

Patterns Identified

- **Pattern 1 (Subject: Action):** Works when action is concise
 - **Pattern 2 (Parentheses):** Excellent for commands/features
 - **Pattern 3 (Simple Extraction):** Needs improvement for long claims
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ALGORITHM ANALYSIS

Current Strengths

1. Preserves parentheses content well
2. Handles technical acronyms (AI, API, PDF)
3. Extracts subject-action relationships
4. Word-boundary truncation prevents mid-word cuts

Current Weaknesses

1. **Length Limit Too Aggressive:** 75 chars may be too short for complex claims
 2. **No Prioritization:** Doesn't prioritize key information (metrics, outcomes)
 3. **Single Feature Extraction:** Loses multiple important features
 4. **No Semantic Understanding:** Doesn't understand what's most important in a claim
 5. **Weak Outcome Handling:** "improved", "created", "added" are too vague
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RECOMMENDATIONS FOR ITERATION 2

High Priority

1. **Increase Length Limit:** Consider 90-100 chars for titles (with better CSS handling)
2. **Metric Preservation:** Prioritize numbers, percentages, and quantitative data
3. **Multiple Features:** Use "and" detection to include multiple features
4. **Outcome Specificity:** Replace vague verbs with more specific descriptions

Medium Priority

1. **Semantic Extraction:** Identify key nouns (what) and key outcomes (result)
2. **Condition Preservation:** Keep important conditions ("when tests pass")
3. **Grammatical Flow:** Improve phrasing for multi-line titles

Low Priority

1. **Context Awareness:** Understand domain-specific important terms
 2. **Abbreviation Handling:** Better handling of technical abbreviations
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NEXT ITERATION PLAN

Starting Conditions (Same as Cycle 1)

- Same 12 test variations
- Same algorithm structure
- Focus on addressing identified weaknesses

Target Improvements

1. ☒ Preserve metrics and percentages
2. ☒ Handle multiple features better
3. ☒ Increase effective title length
4. ☒ Improve outcome specificity
5. ☒ Better truncation strategy

Success Metrics

- All 12 variations should have informative titles
 - No key information lost in truncation
 - Titles should be grammatically correct
 - Titles should fit within display constraints
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TECHNICAL NOTES

Current Implementation

- **File:** scripts/prove_it_comprehensive.py
- **Function:** generate_informative_title()
- **CSS:** src/waft/templates/brief.py (.cover-title class)
- **Length Limit:** 75 characters (with word-boundary truncation)

CSS Improvements Made

- Line-height: 1.2 → 1.4
 - Added word-wrap and overflow-wrap
 - Added padding for multi-line titles
 - Better vertical spacing
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End of Observations - Ready for Iteration 2